

Probability - Set 1

TMUA practice questions by topic. Specification M7.1-M7.7. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 A bag contains 4 red and 6 blue counters. Two counters are taken at random without replacement. Find the probability that they are different colours.

- A $\frac{24}{100}$
B $\frac{15}{25}$
C $\frac{5}{8}$
D $\frac{14}{15}$
E $\frac{12}{25}$
F $\frac{12}{25}$

- 2 Two fair dice are thrown. Find the probability that the product of the two scores is odd.

- A $\frac{2}{3}$
B $\frac{4}{9}$
C $\frac{4}{9}$
D $\frac{6}{9}$
E $\frac{2}{3}$
F $\frac{2}{3}$

- 3 $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.3$. Which of the following statements is/are true?

- I A and B are independent.
II A and B are mutually exclusive.
III $P(A \text{ or } B) = 0.8$

- A none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III

- 4 A spinner has probability 0.15 of landing on red. It is spun 200 times. What is the expected number of times it lands on red?
- A 30
B 40
C 15
D 35
E 25
F 20
- 5 In a class of 30, 18 study French, 12 study German and 5 study both. A student is chosen at random. Given that the student studies French, what is the probability that they also study German?
- A $\frac{1}{6}$
B $\frac{2}{5}$
C $\frac{1}{8}$
D $\frac{1}{9}$
E $\frac{1}{8}$
F $\frac{1}{12}$
- 6 Three fair coins are tossed. Find the probability of getting more heads than tails.
- A $\frac{3}{4}$
B $\frac{1}{4}$
C $\frac{1}{8}$
D $\frac{3}{8}$
E $\frac{1}{8}$
F $\frac{1}{2}$

- 7 The probabilities that a train is early, on time or late are 0.1, x and $2x$ respectively. Find the probability that the train is not late.
- A** 0.5
B 0.9
C 0.7
D 0.4
E 0.6
F 0.3
- 8 A fair die is thrown repeatedly until a six appears. Find the probability that the first six appears on the third throw.
- A** $\frac{25}{216}$
B $\frac{36}{216}$
C $\frac{216}{216}$
D $\frac{25}{216}$
E $\frac{216}{216}$
F $\frac{216}{216}$
- 9 Two cards are drawn at random from a standard pack of 52 without replacement. Which of the following expressions is the probability that both are aces?
- A** $\frac{4}{52} + \frac{3}{51}$
B $\frac{4}{52} \times \frac{3}{52}$
C $\frac{4}{52} \times \frac{3}{51}$
D $\frac{4}{52} \times \frac{3}{52}$
E $\frac{4}{52} \times \frac{3}{51}$
F $\frac{4}{52} \times \frac{3}{52}$

- 10** Events A and B are mutually exclusive with $P(A) = 0.3$ and $P(B) = 0.4$. Find $P(\text{neither } A \text{ nor } B)$.
- A** 0.3
B 0.7
C 0.1
D 0.58
E 0.12
F 0.42
- 11** A biased coin has probability p of landing heads. It is tossed twice. The probability of exactly one head is $\frac{3}{8}$. Find the possible values of p .
- A** $\frac{1}{4}$ or $\frac{2}{3}$
B $\frac{1}{4}$ only
C $\frac{1}{8}$ or $\frac{7}{8}$
D $\frac{3}{4}$ or $\frac{5}{4}$
E $\frac{3}{8}$ only
F $\frac{3}{16}$ only
- 12** A box contains n white balls and 3 black balls. Two balls are drawn without replacement. The probability that both are white is $\frac{5}{12}$. Find n .
- A** 6
B 4
C 5
D 3
E 7
F 9